CHP-7 Current and Ohm’s Law

General Physics 2

2026-10-21

Electric Current

Drude model

In the previous chapters, we studied charges at rest. We now consider electric currents: the transport of electric charge.

The model we will discuss in this course is called Drude model of conductivity. It was established before quantum mechanics so it does not correctly describes the physics of the problem. However, at this level it will be enough for our purposes and the end result for the conductivity will be correct. (For a correct explanation please look for Sommerfeld or Kubo models.)

Around 1900s Drude (Paul Ludwig Drude) reasoned that since metals conduct electricity well they must contain free electrons that move through a lattice of ions. These conduction electrons in a copper wire have rapid, disordered motion, with characteristic speeds of order 10^6 m/s.

Without an applied field, there is no preferred direction of motion. Meaning if there is not electric field through the conductor the random motions of electrons will not create any net transport of charge.

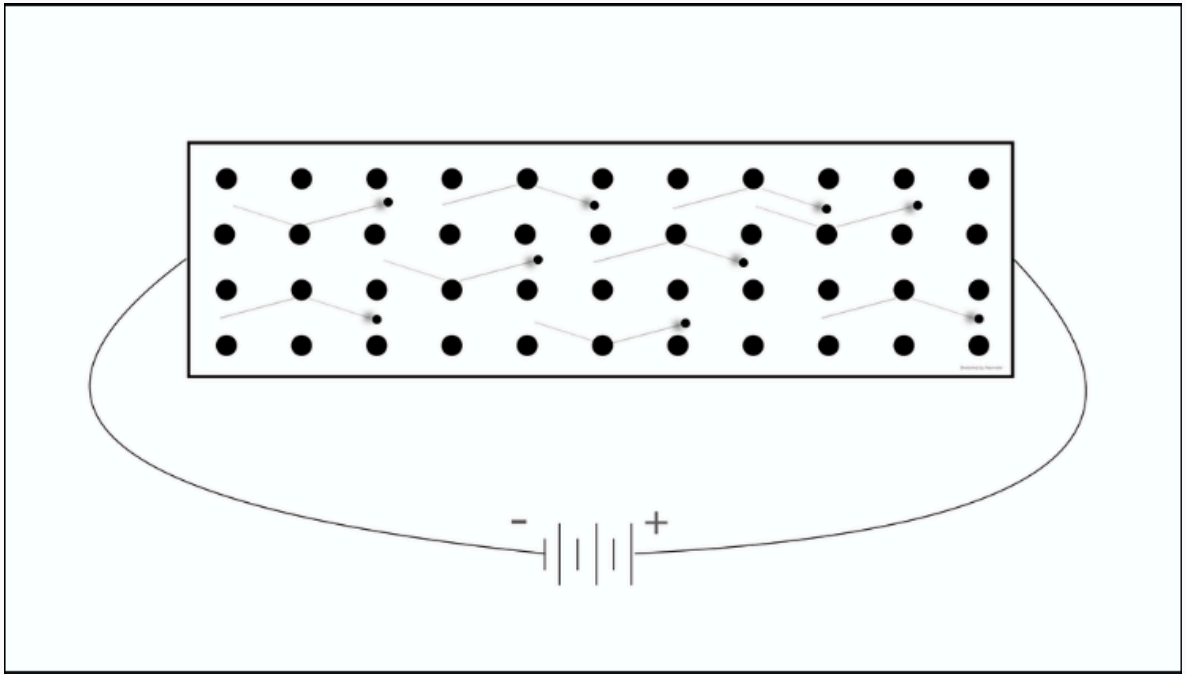


Figure 1: The Drude model.

Establishing a Current

Connecting a wire into a complete circuit with a battery establishes a potential difference between its ends and an electric field inside it.

For a uniform wire of length L, with voltage-drop magnitude ΔV,

$$E = \frac{\Delta V}{L}.$$

The field produces a net drift of the mobile charges. After a short transient, the current can reach a steady state.

$$I = \frac{dq}{dt}, \quad 1\ \text{A} = 1\ \text{C/s}$$

Here dq is the net charge crossing a chosen section in time dt.

Charge Conservation and Current Direction

In steady state, the current is same for all planes which follows from the charge conservation.

- In metals, the mobile charge carriers are electrons.
- In electrolytes, positive and negative ions can carry current.

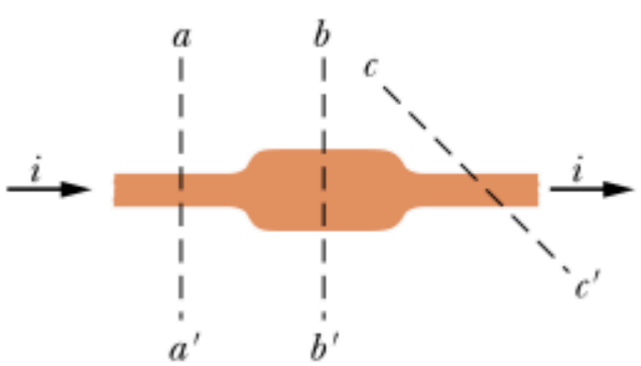
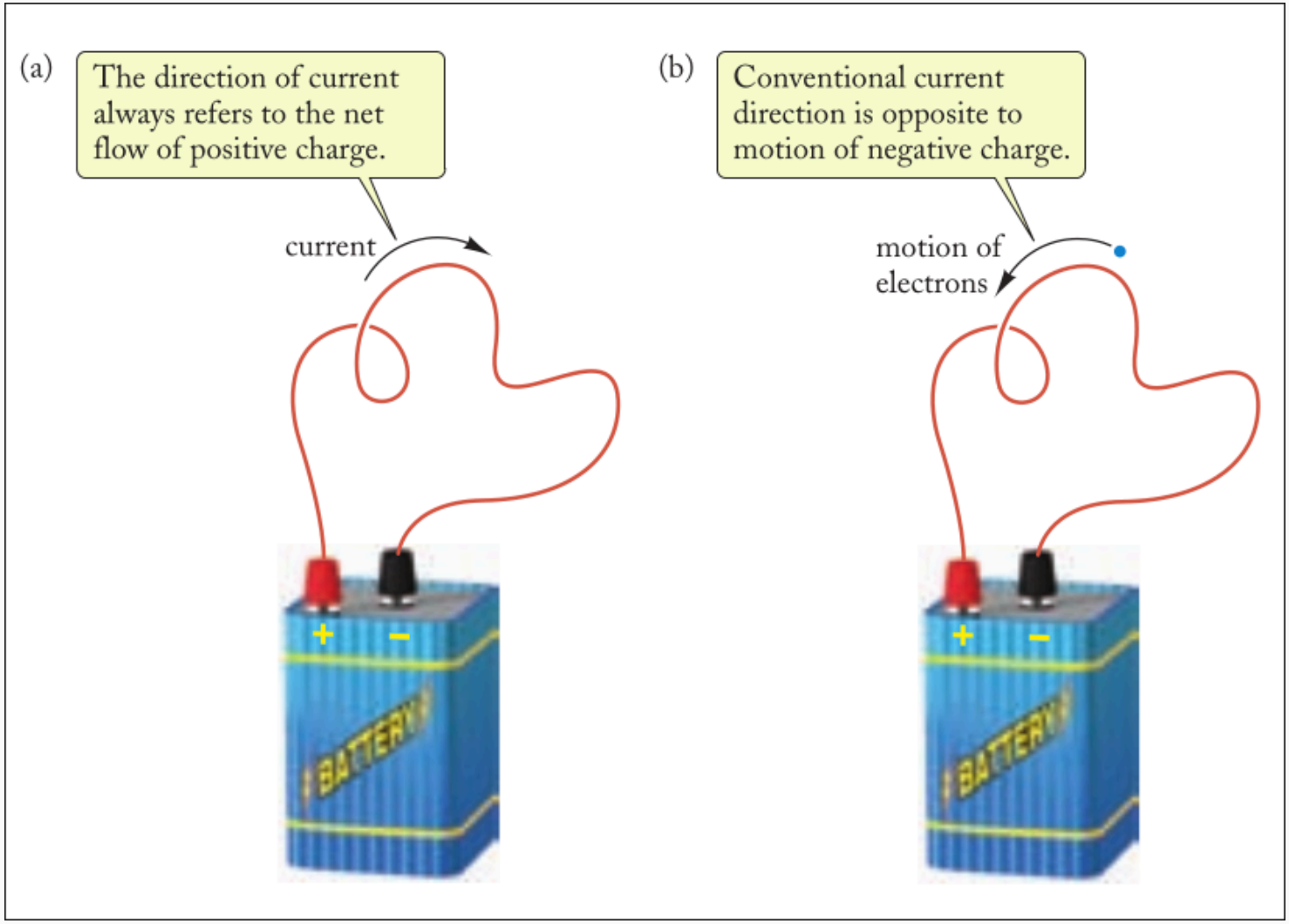


Figure 26.1.2 The current i through the conductor has the same value at planes aa', bb', and cc'.

For historical reasons current direction is chosen in the direction of positive charges, even if the actual charge carriers are negative.


(a)
The direction of current always refers to the net flow of positive charge.
current
+
−
BATTERY
(b)
Conventional current direction is opposite to motion of negative charge.
motion of electrons
+
−
BATTERY

Current Density

The vector current density $\vec{J}$ describes local charge flow. Its flux through an oriented surface gives the current:

$$I = \int_S \vec{J} \cdot d\vec{A}$$

If $\vec{J}$ is uniform and parallel to the surface normal,

$$I = J \int_S dA = JA, \quad J = \frac{I}{A}.$$

The SI unit of current density is A/m². The direction of $\vec{J}$ is the local conventional-current direction.

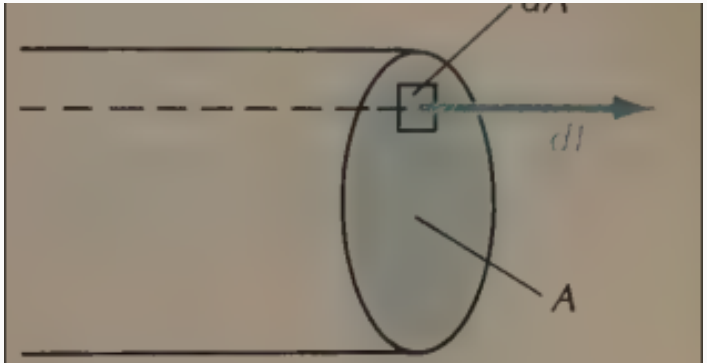


Fig. 28.3 A conductor carries a current from left to right. The conductor has been cut off on the right, so as to give a clear view of its cross section

A Constricted Conductor

Consider a wire that narrows from cross-sectional area A_1 to A_2, where $A_2 < A_1$.

In steady state, charge conservation gives

$$I_1 = I_2 = I.$$

The average current densities across the two sections are

$$J_1 = \frac{I}{A_1}, \quad J_2 = \frac{I}{A_2}.$$

$$\frac{J_2}{J_1} = \frac{A_1}{A_2} > 1$$

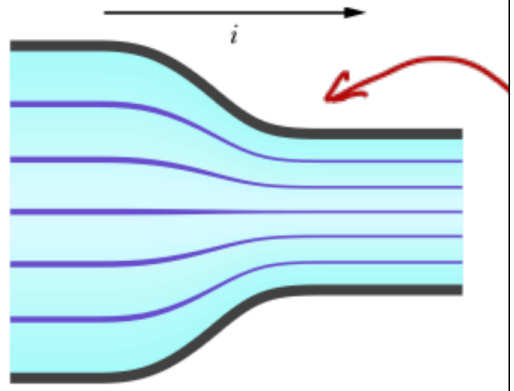


Figure 26.2.1 Streamlines representing current density in the flow of charge through a constricted conductor.

Drift Speed

Random Motion and Drift

With no current, conduction electrons move in all directions with zero average velocity.

With a current, random motion continues, but not they tend to drift with in the opposite direction with a drift velocity $\vec{v}_d$.

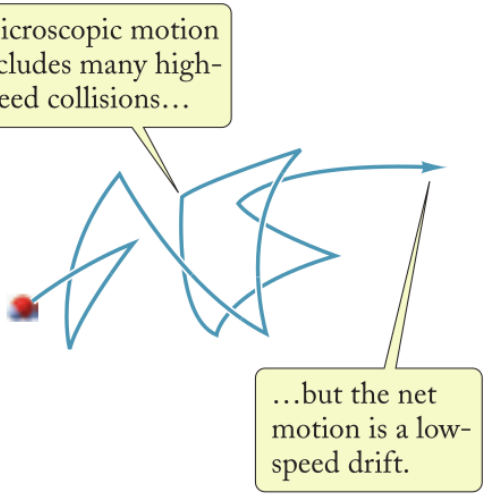


FIGURE 27.6 Path of a water molecule in a canal. The molecule gradually drifts from left to right.

Drift speed is much smaller than the characteristic microscopic speed. It depends on the current, carrier density, and wire area; it is not a fixed property of a metal.

Scattering and a Steady Drift

Electrons accelerate under the electric force between scattering events. Scattering transfers momentum and energy to the material.

Lattice vibrations and imperfections contribute to electrical resistance. In the simple collision model, a characteristic scattering rate for copper is of order $10^{14} s^{-1}$.

Under steady conditions, momentum gained from the field is balanced on average by momentum lost through scattering.

The **average drift velocity** becomes constant. Energy supplied by the field is continually transferred to the material as heat.

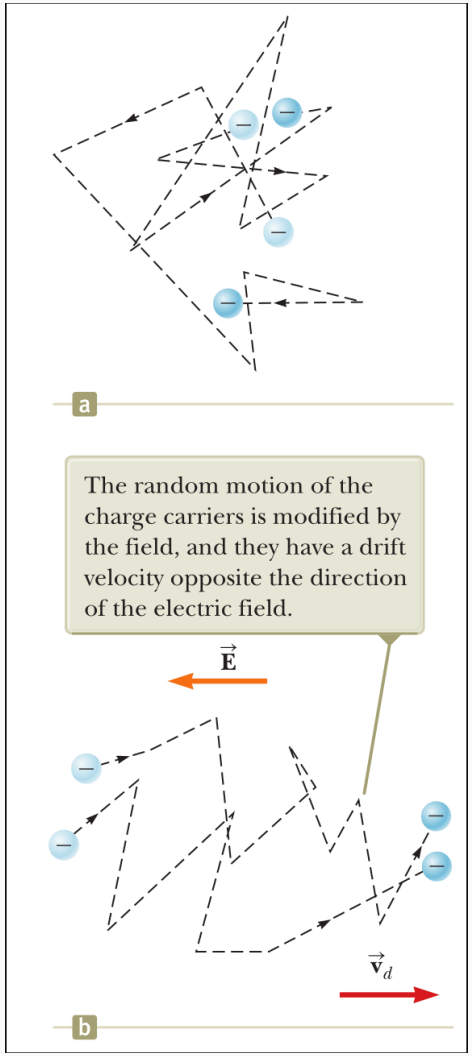


Figure 27.3 (a) A schematic diagram of the random motion of two charge carriers in a conductor in the absence of an electric field. The drift velocity is zero. (b) The motion of the charge carriers in a conductor in the presence of an electric field. Because of the acceleration of the charge carriers due to the electric force, the paths are actually parabolic. The drift speed, however, is much smaller than the average speed, so the parabolic shape is not visible on this scale.

Relating Current to Drift Speed

Let n be the number of mobile carriers per unit volume and A the wire's cross-sectional area. Assume uniform flow with drift speed v_d.

A segment of length L contains nAL carriers. For electrons, its total carrier-charge magnitude is

$$Q = nALe.$$

The drift transit time over this length is $t = \frac{L}{v_d}$. Therefore,

$$I = \frac{Q}{t} = \frac{nALe}{\frac{L}{v_d}} = neAv_d.$$

$$\boxed{v_d = \frac{I}{neA} = \frac{J}{ne}}$$

The Vector Relation and the Charge Sign

For carriers of signed charge q_c,

$$\vec{J} = n q_c \vec{v}_d$$

The carrier charge density nq_c has units $\mathrm{C/m^3}$.

- For positive carriers, $\vec{J}$ and $\vec{v}_d$ point in the same direction.
- For negative carriers, they point in opposite directions.

In particular, for electrons with $e > 0$,

$$\vec{J} = -ne\vec{v}_d, \quad J = nev_d.$$

Example: Drift Speed in a Copper Wire

Example: A copper wire has radius $r = 900\ \mu\text{m}$ and carries a uniform current $I = 17\ \text{mA}$.

What is the drift speed of its conduction electrons?

Solution: Assume one conduction electron per copper atom and uniform current density.

Use the following data:

$$\rho_{\text{mass}} = 8.96 \times 10^{3} \text{kg/m}^3, \quad M = 63.54 \times 10^{-3}\ \text{kg/mol},$$

$$N_A = 6.02 \times 10^{23} \text{mol}^{-1}, \quad e = 1.60 \times 10^{-19}\ \text{C}.$$

Number Density of Conduction Electrons

One conduction electron per atom means the electron number density equals the atom number density:

$$n = \frac{N_A \rho_{\text{mass}}}{M}.$$

Thus,

$$n = \frac{(6.02 \times 10^{23})(8.96 \times 10^{3})}{63.54 \times 10^{-3}} \text{m}^{-3}.$$

$$n \approx 8.49 \times 10^{28} \text{m}^{-3}$$

Here ρ_{mass} is mass density; later ρ will denote resistivity.

Copper-Wire Drift Speed: Result

The cross-sectional area is

$$A = \pi r^2 = \pi\left(900 \times 10^{-6}\ \text{m}\right)^2 \approx 2.54 \times 10^{-6}\text{m}^2.$$

Therefore,

$$v_d = \frac{I}{neA} = \frac{17 \times 10^{-3}}{(8.49 \times 10^{28})(1.60 \times 10^{-19})(2.54 \times 10^{-6})}\ \text{m/s}.$$

$$v_d \approx 4.9 \times 10^{-7}\ \text{m/s}$$

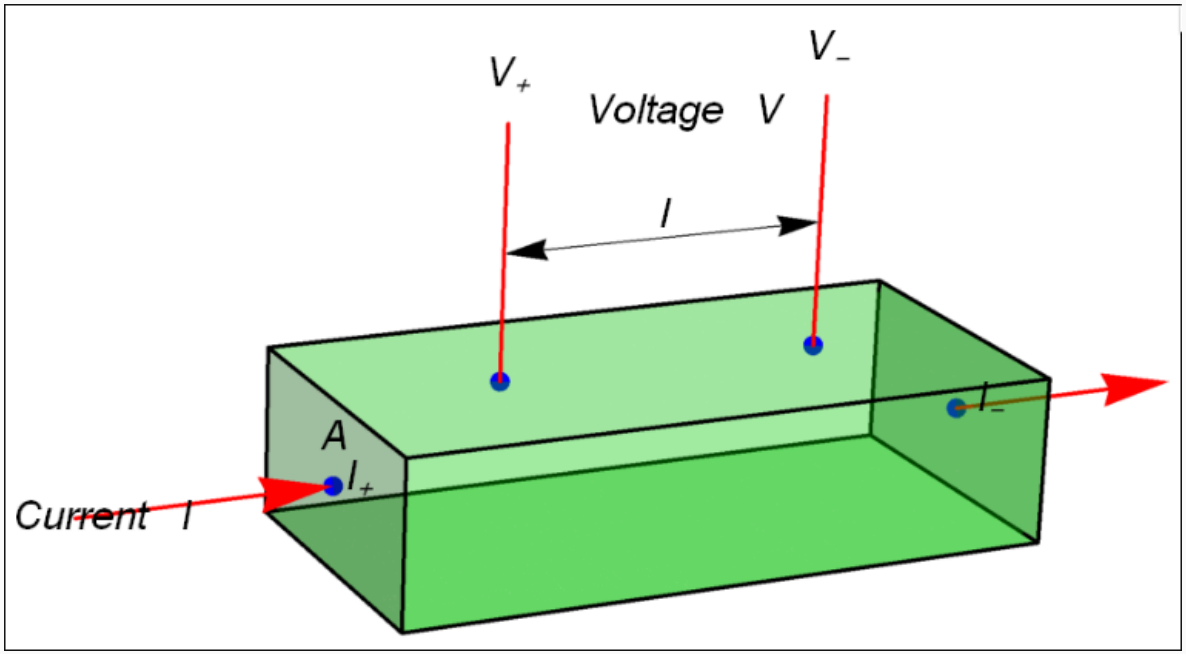
The electron drift direction is opposite to the conventional current.

Resistance and Resistivity

Resistance

Apply a potential difference V between two terminals of a conductor and measure the current I through it. Its resistance at that operating point is

$$R = \frac{V}{I}$$

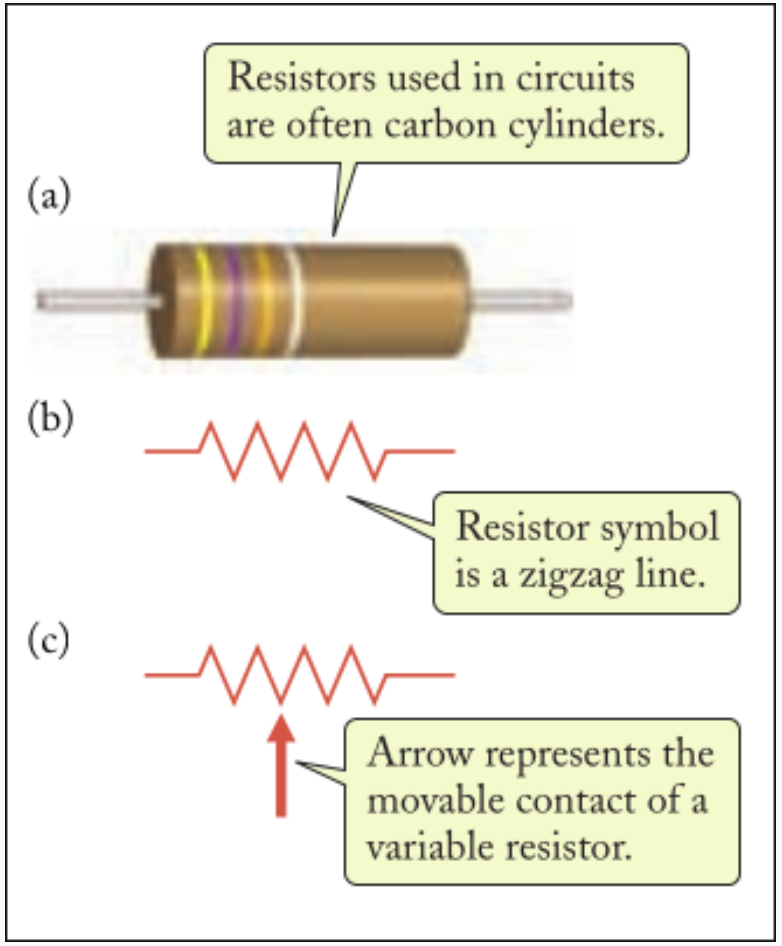


The SI unit is the ohm:

$$1\ \Omega = 1\ \text{V/A}.$$

A component designed to provide a specified resistance is called a resistor.

Its circuit symbol is commonly a zigzag line or a rectangle between two leads.



Resistivity and Conductivity

We will turn back to relation $V = IR$ later but lets focus on resistive property of materials. For an isotropic material, the relation instead of $V = IR$ we have

$$\vec{E} = \rho \vec{J}, \quad \rho = \frac{E}{J}$$

The unit of resistivity is

$$\frac{\mathrm{V/m}}{\mathrm{A/m^2}} = \Omega \cdot \mathrm{m}.$$

Conductivity is the reciprocal of resistivity:

$$\sigma = \frac{1}{\rho}, \quad \vec{J} = \sigma \vec{E}.$$

Conductivity has units S/m, ρ is independent of the applied field.

Calculating Resistance from Resistivity

Consider a uniform rod of length L and cross-sectional area A, with contacts covering both end faces.

If the field and current density are uniform,

$$E = \frac{V}{L}, \quad J = \frac{I}{A}, \quad \rho = \frac{E}{J} = \frac{VA}{LI}.$$

$$R = \frac{V}{I} = \rho \frac{L}{A}$$

V, I, and R describe the whole conductor; E, J, and ρ describe local behavior and material properties.

Resistivity and Temperature

Over a suitable temperature interval, the resistivity of many metals is approximately linear in temperature:

$$\rho = \rho_0[1 + \alpha(T - T_0)]$$

- T_0: reference temperature.
- ρ_0: resistivity at T_0.
- α: temperature coefficient of resistivity, in K^{-1}.

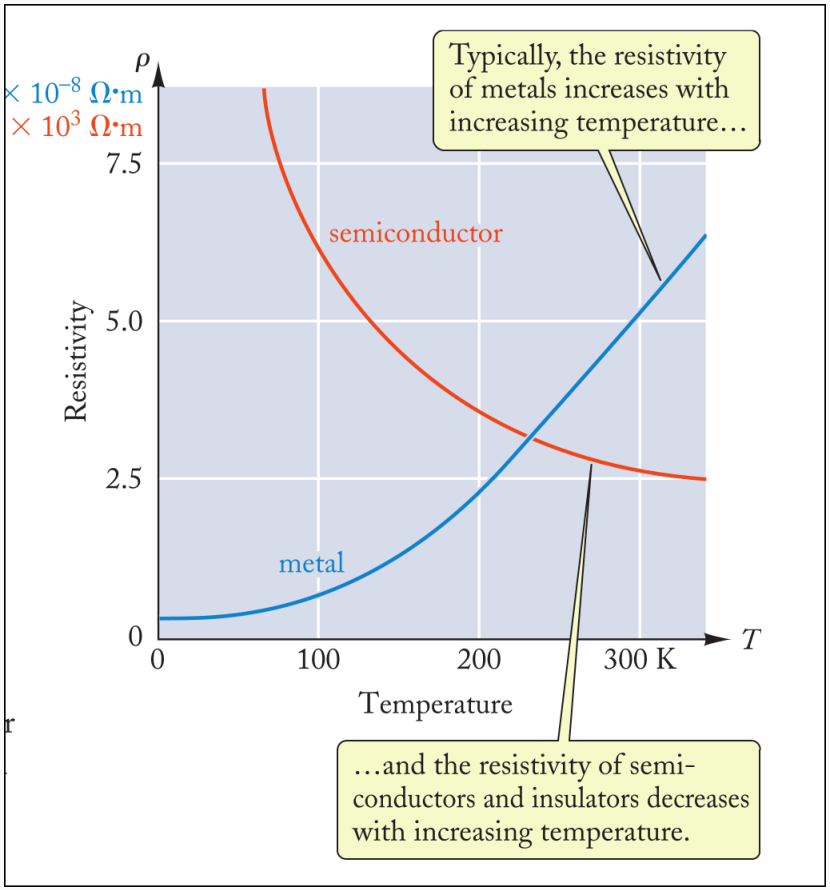
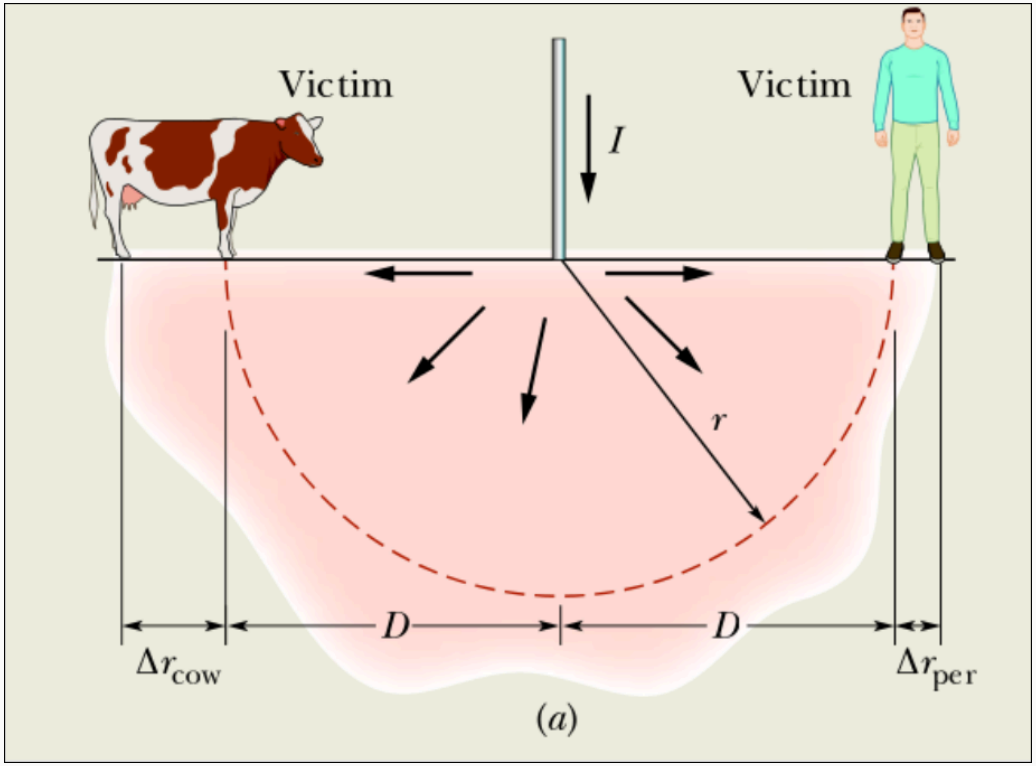


Figure 26.3.5*a* shows a person and a cow, each a radial distance $D = 60.0\,\text{m}$ from the point where lightning of current $I = 100\,\text{kA}$ strikes the ground. The current spreads through the ground uniformly over a hemisphere centered on the strike point. The person's feet are separated by radial distance $\Delta r_{\text{per}} = 0.50\,\text{m}$; the cow's front and rear hooves are separated by radial distance $\Delta r_{\text{cow}} = 1.50\,\text{m}$. The resistivity of the ground is $\rho_{gr} = 100\,\Omega \cdot \text{m}$. The resistance both across the person, between left and right feet, and across the cow, between front and rear hooves, is $R = 4.00\,\text{k}\Omega$.

(a) What is the current i_p through the person?



Ohm’s Law

What Ohm's Law Says

An ohmic device has current directly proportional to applied voltage, when temperature and other physical conditions are held fixed:

$$V = IR, \quad R = \text{constant}$$

Reversing the voltage reverses the current, with the same resistance.

The ratio $R = \frac{V}{I}$ defines a resistance at an operating point even for a non-ohmic device, when the ratio exists.

The additional content of Ohm's law is that R is independent of the applied voltage over the range being considered.

Reading Current-Voltage Characteristics

For an ohmic resistor, the graph of I against V is a straight line through the origin. Its slope is $\frac{1}{R}$.

A semiconductor junction diode has a strongly nonlinear, polarity-dependent characteristic and does not obey Ohm's law.

Many homogeneous materials show an approximately ohmic regime at sufficiently weak fields; strong fields or heating can change their response.

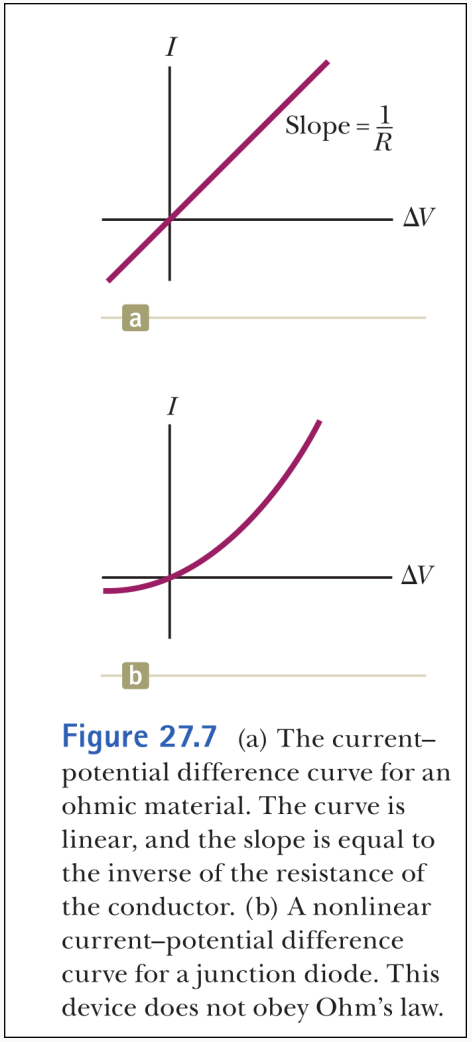
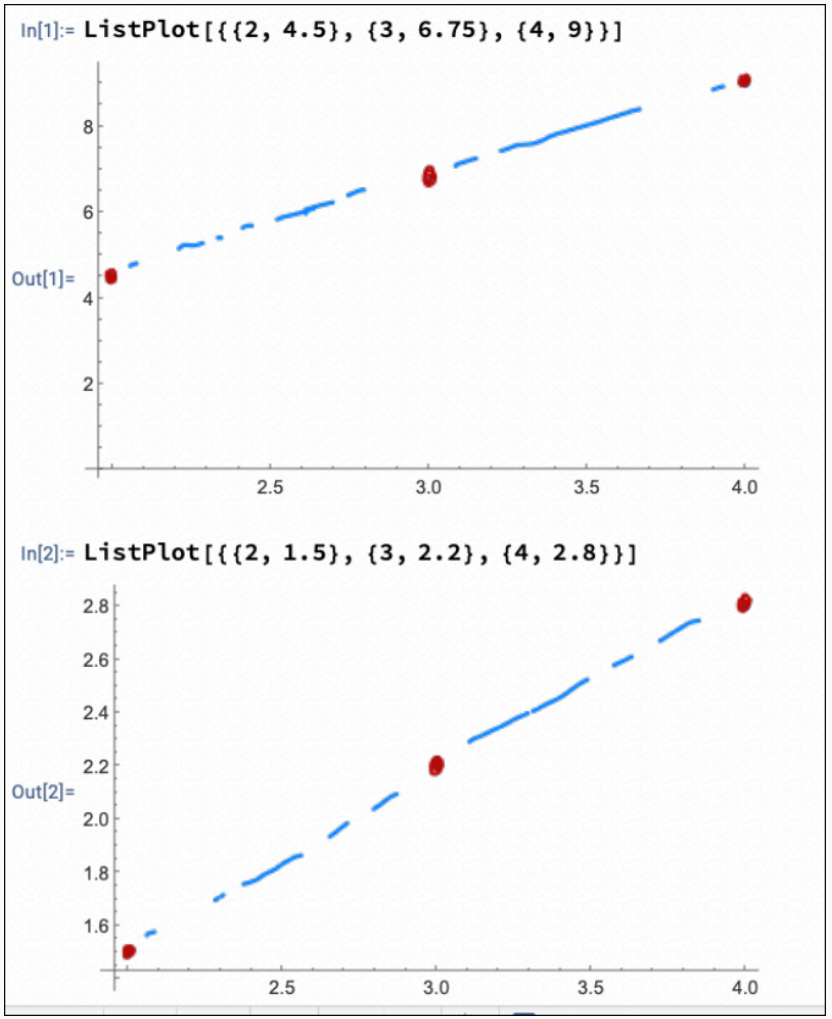


Figure 27.7 (a) The current–potential difference curve for an ohmic material. The curve is linear, and the slope is equal to the inverse of the resistance of the conductor. (b) A nonlinear current–potential difference curve for a junction diode. This device does not obey Ohm's law.

Checkpoint 26.4.1

The following table gives the current i (in amperes) through two devices for several values of potential difference V (in volts). From these data, determine which device does not obey Ohm's law.

Device 1		Device 2	
V	i	V	i
2.00	4.50	2.00	1.50
3.00	6.75	3.00	2.20
4.00	9.00	4.00	2.80



A Microscopic View: The Relaxation Time

Let τ be the average time between the collisions of charge carriers and is a constant that only depends on the material. The Drude model gives us the rate of change of momentum as

$$m\frac{d\vec{v}_d}{dt} = -e\vec{E} - \frac{m\vec{v}_d}{\tau}.$$

The electric force drives the electrons, while the last term represents average momentum loss through scattering.

In steady state,

$$\vec{v}_d = -\frac{e\tau}{m}\vec{E}$$

At fixed material conditions and weak fields, τ is treated as approximately independent of the field. It can change with temperature and impurities.

Microscopic Expression for Resistivity

For electrons, $\vec{J} = -ne\vec{v}_d$. Substituting the steady drift gives

$$\vec{J} = \frac{ne^2\tau}{m}\vec{E}.$$

Comparing with $\vec{E} = \rho\vec{J}$,

$$\rho = \frac{m}{ne^2\tau}, \quad \sigma = \frac{ne^2\tau}{m}$$

The model explains an ohmic response when n and τ stay approximately constant as the applied field changes.

This is consistent with the weak-drift regime $v_d \ll v_{\text{random}}$; it is a model assumption, not a universal field-independent law.

Example: How Slowly Do Electrons Drift?

Example: A lamp is connected by a copper wire of diameter 0.20 cm and length $L = 2.0\ \text{m}$. Take a current of $I = 1.5\ \text{A}$.

For a **steady, one-way current** of this magnitude, estimate the time associated with drifting a distance L.

Use the following values:

$$\rho = 1.7 \times 10^{-8}\ \Omega\cdot\text{m}, \quad \tau = 2.7 \times 10^{-14}\ \text{s},$$

$$e = 1.60 \times 10^{-19}\ \text{C}, \quad m = 9.11 \times 10^{-31}\ \text{kg}.$$

First find $E = \rho \frac{I}{A}$, then $v_d = eE\frac{\tau}{m}$.

Field and Drift Speed in the Wire

The radius is $r = 1.0 \times 10^{-3}\ \text{m}$, so

$$A = \pi r^2 \approx 3.14 \times 10^{-6} \text{m}^2.$$

The field magnitude is

$$E = \rho \frac{I}{A} = \frac{(1.7 \times 10^{-8})(1.5)}{\pi (1.0 \times 10^{-3})^2}\ \text{V/m} \approx 8.1 \times 10^{-3}\ \text{V/m}.$$

Using the supplied relaxation time,

$$v_d = \frac{eE\tau}{m} \approx 3.9 \times 10^{-5}\ \text{m/s}.$$

Drift Time and the AC Qualification

The steady-current drift time is

$$t = \frac{L}{v_d} \approx 5.2 \times 10^4 \text{ s} \approx 14.4 \text{ hours}$$

This is the time scale $\frac{L}{v_d}$ for net drift; microscopic trajectories include rapid disordered motion and scattering.

Power in Electric Circuits

Electrical Energy Transfer

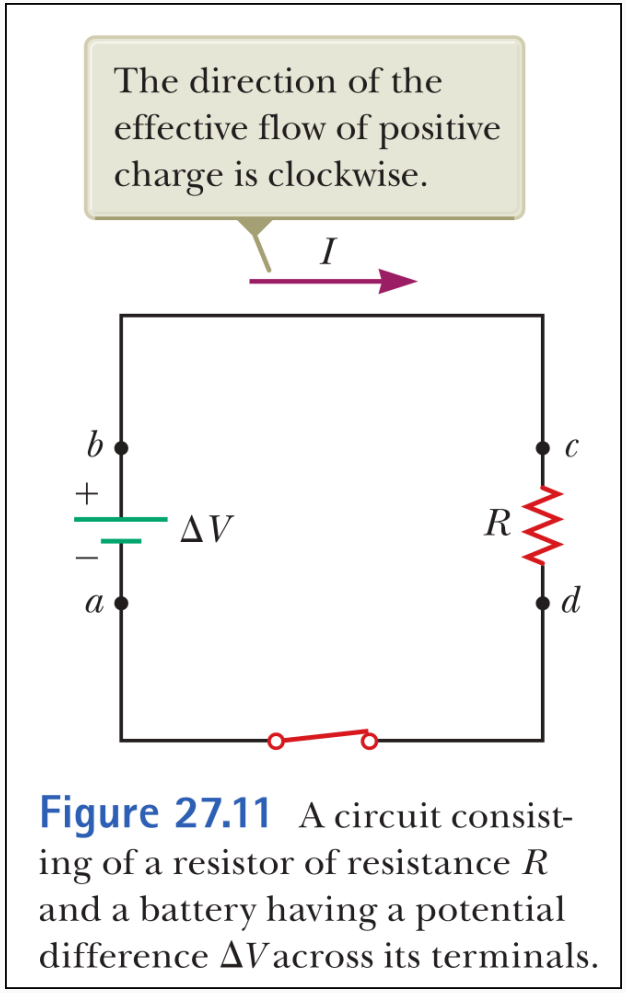


Figure 27.11 A circuit consisting of a resistor of resistance R and a battery having a potential difference ΔV across its terminals.

When charge dq passes through the device, the electrical energy transferred to it is

$$dU = V\,dq = VI\,dt.$$

$$P = \frac{dU}{dt} = IV$$

A motor can convert this energy into mechanical work; a resistor converts it into thermal energy. The unit of power is the watt: 1 W = 1 J/s.

Power Dissipated in a Resistor

For a resistor obeying $V = IR$, substitute into $P = IV$:

$$P = I^2R = \frac{V^2}{R}$$

These expressions describe resistive heating.

With voltage and current defined consistently at a two-terminal device, the instantaneous transferred power is $P = IV$. The resistor formulas follow by using its resistive voltage-current relation.

For time-dependent currents, average power must be obtained from the time average of $I(t)V(t)$.

Superconductors

Superconductivity

In 1911, Heike Kamerlingh Onnes discovered that mercury loses its electrical resistance below a transition temperature of about 4.2 K (-268 C).

This phenomenon is called superconductivity. A superconductor supports current with zero DC resistance under appropriate conditions.

In 1986, superconductivity was discovered in a class of ceramic materials at substantially higher temperatures.

Superconductivity is limited by critical temperature, magnetic field, and current conditions.

Electron Pairing

Conventional superconductors are described by BCS theory. Electrons form correlated Cooper pairs that participate in a collective quantum state.

The conventional pairing mechanism does not describe every class of superconductor in the same way.

Table 27.3 **Critical Temperatures for Various Superconductors**

Material	T_c (K)
$HgBa_2Ca_2Cu_3O_8$	134
Tl—Ba—Ca—Cu—O	125
Bi—Sr—Ca—Cu—O	105
$YBa_2Cu_3O_7$	92
Nb_3Ge	23.2
Nb_3Sn	18.05
Nb	9.46
Pb	7.18
Hg	4.15
Sn	3.72
Al	1.19
Zn	0.88